F79SU – SOLUTIONS – 2024-25

Maps to IFoA subject CS2 – *Atuarial statistics 2 (Risk Modelling and Survival Analysis).* Weighting of IFoA subject: 25%.

1. *Parts (a) and (b) are similar to examples shown in class and should be straightforward given the hints. Part (c) will test students' understanding on the suitability of different hazard rates for mortality.*
CS2 syllabus topic 4.1 (Concepts of survival models).

(a) The survival function is given as

$$S(x) = 1 - F(x) = \exp(-\lambda x)$$

and therefore the hazard rate is

$$\mu_x = \frac{f(x)}{S(x)} = \frac{\lambda \exp(-\lambda x)}{\exp(-\lambda x)} = \lambda.$$

[2]

(b) We now have:

$$\begin{aligned} S(x) &= \int_x^\infty \nu\lambda y^{\nu-1} \exp(-\lambda y^\nu) dy \\ &= [-\exp(-\lambda y^\nu)]_x^\infty = \exp(-\lambda x^\nu), \quad \lambda, \nu > 0. \end{aligned}$$

and therefore the hazard rate is given by

$$\mu_x = \frac{f(x)}{S(x)} = \frac{\nu\lambda x^{\nu-1} \exp(-\lambda x^\nu)}{\exp(-\lambda x^\nu)} = \nu\lambda x^{\nu-1}.$$

[3]

(c) With T having an exponential distribution in (a), the hazard rate is constant (λ) with time, so this may not be suitable for describing the lifetime in mortality studies.

In (b), with T having a Weibull distribution, for any $\lambda > 0$, the hazard rate depends on the value of parameter ν:
if $\nu = 1$, or $\nu < 1$, the hazard rate will be constant or decreasing with time respectively – which again may not be suitable for the lifetime in mortality studies;
if $\nu > 1$, the hazard rate will be increasing with time – this is more suitable for describing lifetime. [4]

[Total 9]

2. *This is similar to class examples and tutorial-type questions.*
CS2 syllabus topic 4.2 (Estimation of lifetime distributions)

Further treatment is required at times 6, 8, 11, 13, 15, 18 and 19, so the partial likelihood is

$$\begin{aligned} L(\beta) &\propto \frac{e^\beta}{8+8e^\beta}\cdot\frac{1}{7+7e^\beta}\cdot\frac{1}{6+6e^\beta}\cdot\frac{1}{3+6e^\beta}\cdot\frac{e^\beta}{2+4e^\beta}\cdot\frac{1}{2+2e^\beta}\cdot\frac{1}{1+2e^\beta} \\ &\propto \frac{e^{2\beta}}{(1+e^\beta)^4(1+2e^\beta)^3} \end{aligned}$$

with partial log likelihood

$$\ell(\beta) = 2\beta - 4\log(1+e^\beta) - 3\log(1+2e^\beta).$$

[3]

(b)

$$U(\beta) = 2 - \frac{4e^\beta}{1+e^\beta} - \frac{6e^\beta}{1+2e^\beta}.$$

The MLE of $x = e^\beta$ satisfies

$$\begin{aligned} 2(1+x)(1+2x) - 4x(1+2x) - 6x(1+x) &= 0 \\ \Rightarrow 10x^2 + 4x - 2 &= 0 \end{aligned}$$

and so $\hat{x} = 0.2899$ and $\hat{\beta} = \log\hat{x} = -1.238$.

[3]

(c) The new drug appears to be more effective. [1]

$e^{\hat{\beta}} < 1$ which means that the hazard rate is lower for the new drug. [1]

(d) We have $\ell(\hat{\beta}) = -4.8666$ and $\ell(0) = -6.0684$

hence $-2\log\Lambda = -2(\ell_0 - \ell_1) = 2.4036$. [2]

With $\chi^2_1(0.05) = 3.841$, we do not have enough evidence to reject H_0 (even though the new drug appeared to perform better). [1]

[Total 11]

3. *Similar to to examples shown in class and to tutorial questions. Part b(ii) will require combining knowledge from different parts of Markov models, and will test further understanding.*
CS2 syllabus topic 4.3 (Derive maximum likelihood estimators for transition intensities).

(a) Suppose the transfers occur in small intervals of time δt_1 and δt_2 at Apr 1 and May 1 respectively. The probability of the given sequence of events is

$$e^{-3\alpha}\left[\alpha\,\delta t_1 + o(\delta t_1)\right] e^{-(\beta+\sigma)}\left[\beta\,\delta t_2 + o(\delta t_2)\right] e^{-4\alpha}.$$

[3]

Dividing through by δt_i and letting $\delta t_i \to 0$, $i = 1, 2$, gives the contribution as

$$L(\alpha, \beta, \sigma, \rho) \propto e^{-7\alpha}\alpha \times e^{-\sigma} \times e^{-\beta}\beta.$$

[2]

(b)

(i) Standard formulae give

$$\begin{aligned}
\hat{\alpha} &= \frac{T_{wm}}{W_w} = \frac{500}{8000} = 0.0625; \quad \hat{\sigma} = \frac{T_{ms}}{W_m} = \frac{100}{1500} = 0.0667; \\
\hat{\beta} &= \frac{T_{mw}}{W_m} = \frac{300}{1500} = 0.2; \quad \hat{\rho} = \frac{T_{sm}}{W_s} = \frac{50}{500} = 0.1
\end{aligned}$$

[2]

Standard errors are

$$\begin{aligned}
\mathrm{SE}(\hat{\alpha}) &= \sqrt{\frac{\hat{\alpha}}{W_w}} = \sqrt{\frac{0.0625}{8000}} \approx 0.00280; \quad \mathrm{SE}(\hat{\sigma}) = \sqrt{\frac{\hat{\sigma}}{W_m}} = \sqrt{\frac{0.0667}{1500}} \approx 0.00667; \\
\mathrm{SE}(\hat{\beta}) &= \sqrt{\frac{\hat{\beta}}{W_m}} = \sqrt{\frac{0.2}{1500}} \approx 0.01155; \quad \mathrm{SE}(\hat{\rho}) = \sqrt{\frac{\hat{\rho}}{W_s}} = \sqrt{\frac{0.1}{500}} \approx 0.01414.
\end{aligned}$$

[2]

(ii)

$$\Pr(\text{Remains well for 12 months}) = e^{-12\alpha} \approx e^{-12*0.0625} = 0.4724$$

[2]

[Total 10]

4. *Similar examples to all parts seen in class and tutorials.*
CS2 syllabus topic 4.3 (ML estimators for transition intensities).

(a) Under the assumption of constant force of mortality we can compute the likelihood contribution for each life as follows:

Life	q_i	d_i	Contribution
1	$1-\exp(-\mu)$	0	$\exp(-\mu)$
2	$1-\exp(-\mu)$	1	$1-\exp(-\mu)$
3	$1-\exp(-0.5\mu)$	1	$1-\exp(-0.5\mu)$
4	$1-\exp(-0.4\mu)$	0	$\exp(-0.4\mu)$
5	$1-\exp(-0.5\mu)$	1	$1-\exp(-0.5\mu)$
6	$1-\exp(-0.6\mu)$	0	$\exp(-0.6\mu)$

[2]

Therefore, the likelihood function is

$$\begin{aligned} L(\mu) &\propto e^{-\mu} \times (1-e^{-\mu}) \times (1-e^{-0.5\mu}) \times e^{-0.4\mu} \times (1-e^{-0.5\mu}) \times e^{-0.6\mu} \\ &= e^{-2\mu}\left(1-e^{-\mu}\right)\left(1-e^{-0.5\mu}\right)^2 \end{aligned}$$

[2]

(b) With $e^{-\mu} = \theta^2$ we have:

$$L(\theta) = \theta^4(1-\theta^2)(1-\theta)^2$$

and

$$\begin{aligned} \ell(\theta) &= 4\log(\theta) + \log(1-\theta^2) + 2\log(1-\theta) \\ &= 4\log(\theta) + \log(1+\theta) + \log(1-\theta) + 2\log(1-\theta) \\ &= 4\log(\theta) + \log(1+\theta) + 3\log(1-\theta) \end{aligned}$$

Setting the first derivative of $\ell(\theta)$ to zero gives:

$$\frac{4}{\theta} - \frac{3}{1-\theta} + \frac{1}{1+\theta} = 0 \Rightarrow 8\theta^2 + 2\theta - 4 = 0$$

and so $\hat{\theta} = 0.593$, and so $\hat{q} = 1 - e^{-\hat{\mu}} = 1 - \hat{\theta}^2 = 0.648$.

[4]

(c) (i) The exact waiting time is $1 + 0.5 + 0.3 + 0.4 + 0.3 + 0.6 = 3.1$.
(ii) There are 3 deaths. The MLE of μ is $3/3.1 = 0.968$ and so the MLE of q is $1 - e^{-0.968} = 0.62$.

[3]
[Total 11]

5. *Similar examples to all parts seen in class and tutorials. Part (c) will test students' ability to critically comment on graduation.*
CS2 syllabus topic 4.5 (Graduation and tests)

(a) We have

$$\text{logit}(\dot{q}_{60}) = 1.19\text{logit}(q^s_{60}) = -1.19 \times 4.312 = -5.131$$

So

$$\frac{\dot{q}_{60}}{(1-\dot{q}_{60})} = e^{-5.131} = 0.00591 \Rightarrow \dot{q}_{60} = \frac{0.00591}{1+0.00591} = 0.005875;$$

It follows that

$$\dot{d}_{60} = 66833 \times 0.005875 = 392.6$$

and

$$z_{60} = (447 - 392.6)/\sqrt{392.6} = 2.75.$$

[4]

(b) Sign test: there are 7 +'s and 4 -'s. This is clearly unexceptional and we have no evidence that the graduation is unacceptable. Formally, we want $2Pr(S \geq 7)$ where $S \sim Bin(11, 1/2)$. Now $2Pr(S \geq 7) = 2Pr(S \leq 4) = 2 \times 0.2744 = 0.549$.

Change of sign test: there are 7 +'s and 4 -'s and 4 changes of sign. This is a one-sided test and we compute $Pr(C \leq 4) = 0.377$ since $C \sim Bin(10, 1/2)$. Again no evidence that the graduation is unacceptable.

Grouping of signs test: we have $n_1 = 7$ positive signs, $n_2 = 4$ negative signs and $g = 3$ groups of positive signs. From tables (e.g. 'yellow' book), the largest value g^* for which we would reject the null hypothesis (that positive and negative residuals are in random order) at the 5% level, is $g^* = 1$. So again there is no evidence that the graduation is unacceptable.

[6]

(c) Although the graduation has passed all three tests comfortably, the residual plot does suggest that the graduation underestimates the death rate at younger ages. [2]

[Total 12]

6. *Similar examples to all parts seen in class and tutorials. Historically, students find the concept of rate intervals and correspondence to estimates challenging, so the question will test their understanding on this topic.*
CS2 syllabus topic 4.4 (Principle of correspondence; Exposed to risk).

(a) We have

$P^C_{x,t}$: number of lives aged x *nearest* birthday at time t;
$P^D_{x,t}$: number of lives aged x *last* birthday at time t.

Assuming births are uniformly distributed throughout the year:

$$P^D_{x,t} \approx \frac{1}{2}(P^C_{x,t} + P^C_{x+1,t}).$$

Therefore:

$$P^D_{80,2013} \approx \frac{1}{2}(P^C_{80,2013} + P^C_{81,2013}) = \frac{1}{2}(489 + 524) = 506.5.$$

[2]

(b)

$$\hat{\mu}_{80} = \frac{d_{80}}{E^c_{80}} = \frac{32}{1516.25} = 0.0211047.$$

Age 80 last birthday means that exact age is in [80,81).
So, estimate $\hat{\mu}_{80}$ applies to age 80.5 (midpoint of the interval).

[2]

(c)

$$\hat{q}_{80} = 1 - e^{-\hat{\mu}_{80}} = 1 - e^{-0.0211047} = 1 - 0.9791164 = 0.02088355.$$

This estimate applies at age 80 (start of the interval). [2]

[Total 6]

[END OF SOLUTIONS]